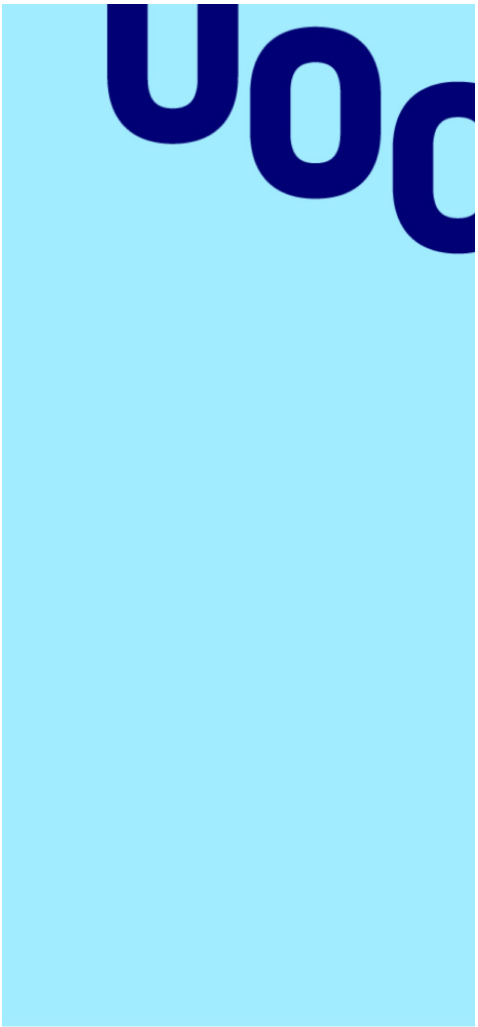
Módulo 1 - Definición y planificación del trabajo final



Universitat Oberta de Catalunya

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Area 4: Pattern Recognition and Computer Vision in Applied Sciences - Geographic Information Systems, satellite images and geopositioning

MSc in Data Science

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Contents

1 Proposed title 3

2 Keywords 4

3 Abstract 5

4 Description of the proposal and justification of its interest and relevance 6

5 Explanation of personal motivation 8

6 Definition of objectives (primary and secondary) 9

7 Description of the methodology used in the development of the project 10

8 Research plan 12

Chapter 1

Proposed title

Comparative Evaluation of Satellite Image Segmentation Methods for Coastline Extraction: Spectral Indices vs. Deep Learning.

Chapter 2

Keywords

Earth Observation, Satellite imagery, Coastal Remote Sensing, Semantic Segmentation, Spectral Indices, Water Segmentation.

Chapter 3

Abstract

Accurate coastline extraction using high-frequency satellite data like Sentinel-2 is a cornerstone of modern coastal engineering, but current automated methods face severe limitations. Environments characterized by large intertidal zones and complex hydrodynamics induce spectral confusion among water, wave foam, and wet sand. This project aims to assess if state-of-the-art Deep Learning models can overcome the shortcomings of classical pixel-based remote sensing techniques.

The methodology revolves around creating a high-quality, AI-annotated Ground Truth dataset covering two morphodynamically distinct locations (Cantabrian and Mediterranean coasts) to train and test advanced semantic segmentation models. These models will be strictly benchmarked against established baselines, which rely on the automatic Otsu thresholding of spectral indices such as the Normalized Difference Water Index (NDWI) and Modified NDWI (MNDWI). To capture both the correct classification of pixels and the physical accuracy of the shoreline, the evaluation will be twofold: utilizing standard computer vision metrics, like Intersection over Union (IoU) and Dice Coefficient, alongside sub-pixel geometric validation (mean Euclidean distance). Ultimately, this research will quantify the performance degradation of each method under adverse conditions and their cross-site transferability, defining the most robust approach for future coastal monitoring frameworks.

Chapter 4

Description of the proposal and justification of its interest and relevance

Coastal regions are among the most dynamic and socioeconomically vital areas on Earth. Accurate shoreline detection is essential for monitoring erosion and planning coastal defense strategies. This task is increasingly being addressed through satellite remote sensing, where missions such as Sentinel-2 offer high-frequency and multi-decadal data for coastal morphological characterization [Vitousek et al., 2023]. Currently, established frameworks like CoastSat [Vos et al., 2019], SHOREX [Sánchez-García et al., 2020], and CASSIE [Almeida et al., 2021] have successfully implemented workflows to define satellite-derived waterlines, providing low-cost information to track shoreline evolution across various scales.

However, these traditional methods based on spectral index thresholds often struggle with "noise," such as wave foam in swash zones and mixed pixels at the water-land interface. This is particularly evident on the Cantabrian coast, where high tides create a large intertidal area in which differences in water saturation in the sand can hinder waterline detection (see Figure 4.1). Consequently, there is a clear need to explore more robust, context-aware approaches like Deep Learning to overcome the spectral confusion inherent in these highly dynamic environments.

This proposal is relevant because it seeks to bridge the gap between classical remote sensing and modern computer vision by providing a quantified comparison of the accuracy gained by using deep learning models versus traditional spectral methods.

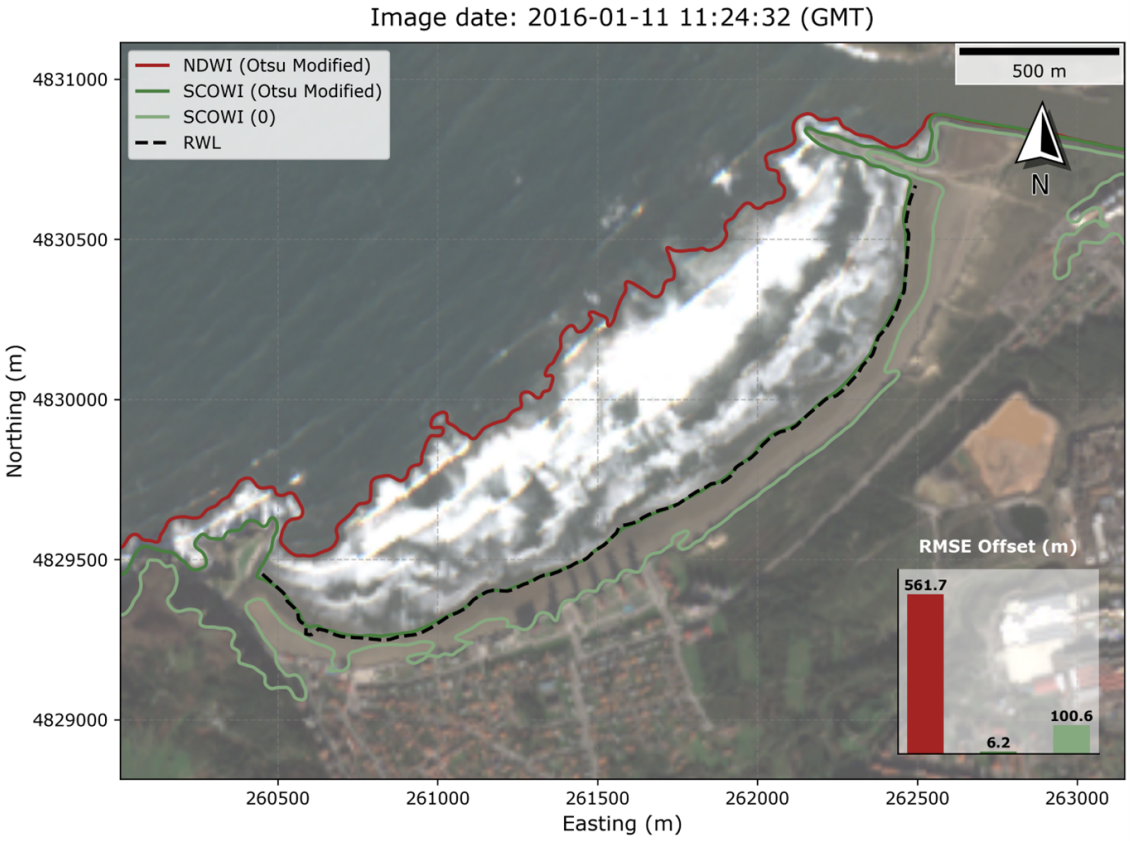


Figure 4.1: Examples of shoreline misidentification in a macrotidal beach (adapted from [Cabezas-Rabadán et al., 2026])

Chapter 5

Explanation of personal motivation

Beyond my academic path in this master's program, I currently work as a Coastal Engineer, specifically focusing on coastal monitoring through remote sensing and satellite imagery. My professional experience has provided me with a deep understanding of coastal dynamics; however, it has also exposed the technical limitations of traditional engineering tools when dealing with highly dynamic environments.

To address this, I aim to implement Deep Learning as it represents a significant scientific advancement over current pixel-based methods. Unlike traditional indices that rely on rigid thresholds, these models learn spatial context and complex textures, allowing them to resolve the spectral confusion between water, foam, and wet sand.

Chapter 6

Definition of objectives (primary and secondary)

Primary Objective:

- To perform a comprehensive benchmarking analysis between traditional spectral-based thresholding and Deep Learning-based semantic segmentation for coastline extraction.

Secondary Objectives:

- To generate a high-quality "Ground Truth" dataset of coastal masks using AI-assisted annotation (SAM) [Kirillov et al., 2023].
- To evaluate the performance degradation of each method under adverse conditions (foam, differences in sand moisture content, and clouds).
- To assess the transferability of trained models between different coastal typologies (Cantabrian vs. Mediterranean).

Chapter 7

Description of the methodology used in the development of the project

The methodology is structured into four main phases as illustrated in the workflow diagram (Figure 7.1):

1. **Data Acquisition:** Downloading Sentinel-2 Level-2A (atmospherically corrected) products from Google Earth Engine, covering the period 2015–2025. A built-in cloud detection algorithm will be applied as an initial filter to discard obstructed scenes, utilizing the remaining available cloud-free images.
2. **Preprocessing & Labelling:** Calculation of spectral indices (NDWI, MNDWI). Generation of Ground Truth masks using the Segment Anything Model (SAM) through an interactive annotation tool [Bhattiprolu, 2026].
3. **Model Implementation:**
   - *Baseline:* Automatic Otsu thresholding applied to spectral indices.
   - *Deep Learning:* Training semantic segmentation models like U-Net [Ronneberger et al., 2015] or SegFormer [Xie et al., 2021]. Model training will be run locally on an NVIDIA RTX 3060 GPU, with an institutional computing cluster (2 nodes, 2x NVIDIA A30 each) available for scaling if required.
4. **Evaluation:** Comparison using pixel-based metrics (IoU, Dice) and geometric validation (calculating the mean Euclidean distance between the predicted and real coastline).

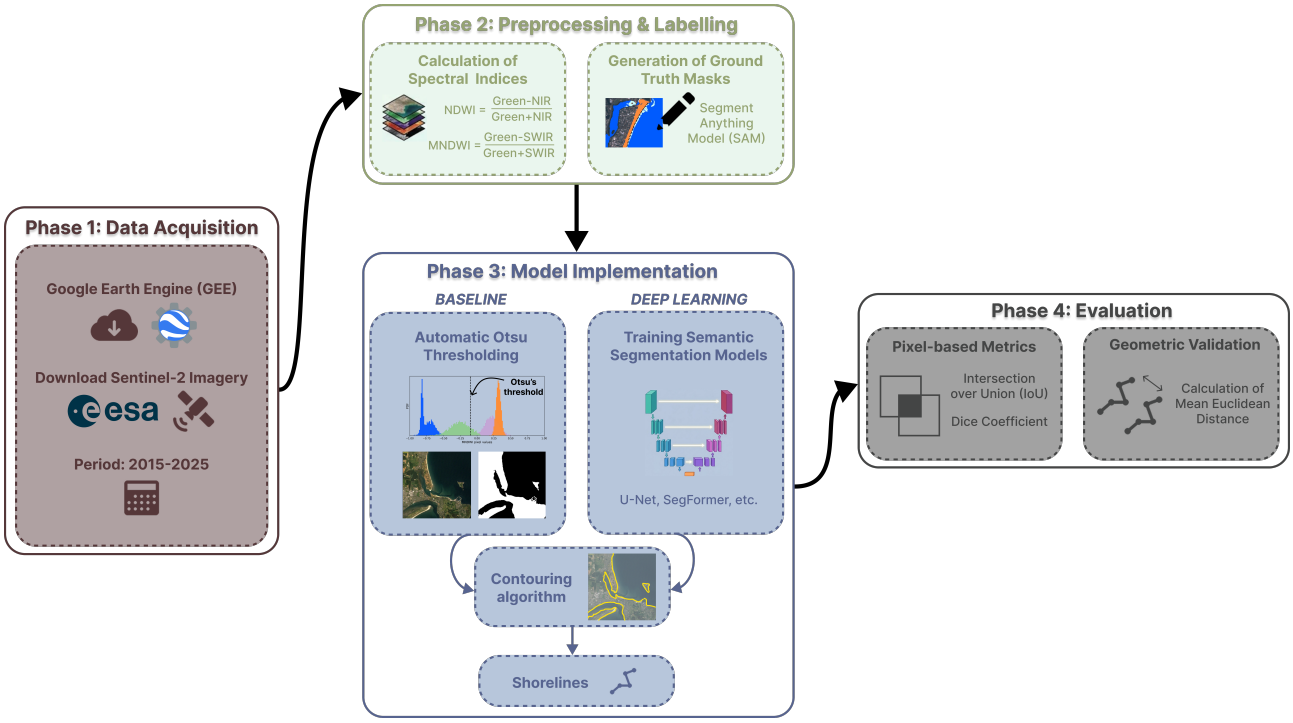


Figure 7.1: Schematic representation of the methodological workflow.

Chapter 8

Research plan

The project is planned to be developed over a 5-month period (including the thesis defence), structured into the following tasks as shown in the Gantt chart (Figure 8.1). Additionally, project management software such as Trello or MS Planner will be used to track and control the tasks at a more specific level:

- **Task 1:** Literature review, selection of study sites (beaches), and satellite images download.
  - Literature review on traditional methods for satellite-derived coastline extraction.
  - Literature review on Deep Learning models applied to coastal monitoring.
  - Satellite images download for Beach 1 (Cantabrian environment).
  - Satellite images download for Beach 2 (Mediterranean environment).
- **Task 2:** Calculation of spectral indices and generation of Ground Truth masks.
  - Preparation of the GitHub repository structure for the project's codebase.
  - Calculation of spectral indices (e.g., NDWI, MNDWI).
  - Create false-color composites to assist in the annotation process.
  - Ground Truth generation through AI-assisted annotation of the dataset.
- **Task 3:** Model implementation: Development of baseline and Deep Learning architectures.
  - Application of automatic thresholding to spectral indices to set baseline models.
  - Selection, development, and training of Deep Learning architectures.
  - Application of contouring algorithms to extract shorelines at sub-pixel precision.
- **Task 4:** Statistical analysis of results, benchmarking across different algorithms, and final evaluation.
  - Pixel-based metrics evaluation (e.g., IoU, Dice Coefficient).
  - Geometric validation by calculating the mean Euclidean distance between predicted and real shorelines.

  - Comparison of performance under adverse conditions (foam, wet sand) and transferability assessment.

- **Task 5:** Drafting report and interim deliveries.
- **Task 6:** Progress meetings with directors.

TASK	Feb				Mar					Apr				May				Jun				
	2	9	16	26	2	9	16	26	30	6	13	20	27	4	11	18	25	1	8	15	22	29
T1: Literature review, selection of study sites and satellite images download.																						
T2: Ground Truth generation and calculation of spectral indices.																						
T3: Model implementation: Development of baseline and Deep Learning architectures.																						
T4: Statistical analysis of results, benchmarking across different algorithms, and final evaluation.																						
T5: Drafting report and interim deliveries.																						
T6: Progress meetings with directors.																						

Figure 8.1: Gantt chart of the project timeline.

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